Master Memory Leaks in React.js

A **memory leak** happens when memory is allocated but never released — objects that should be garbage-collected stay alive because **something still references them**.

In React, leaks happen when components don't clean up resources properly before unmounting — like:

- Async calls still resolving
- Subscriptions that never unsubscribe
- Event listeners never removed
- Timers that keep ticking
- Closures keeping stale references

Common Causes of Memory Leaks in React:

1. **Event Listeners:**
   If you add event listeners to the DOM, window, or document and don't remove them when the component unmounts, memory can get stuck.
2. **setState on Unmounted Components:**
   Calling setState after a component is removed from the screen keeps old data in memory.
3. **Subscriptions and External Resources:**
   Not unsubscribing from things like WebSockets, APIs, or data streams can stop memory from being cleared.
4. **Closures:**
   Functions that "remember" old state or props can accidentally hold memory, even after the component is gone.

Debug a Memory Leak

Use Chrome DevTools → Performance Tab

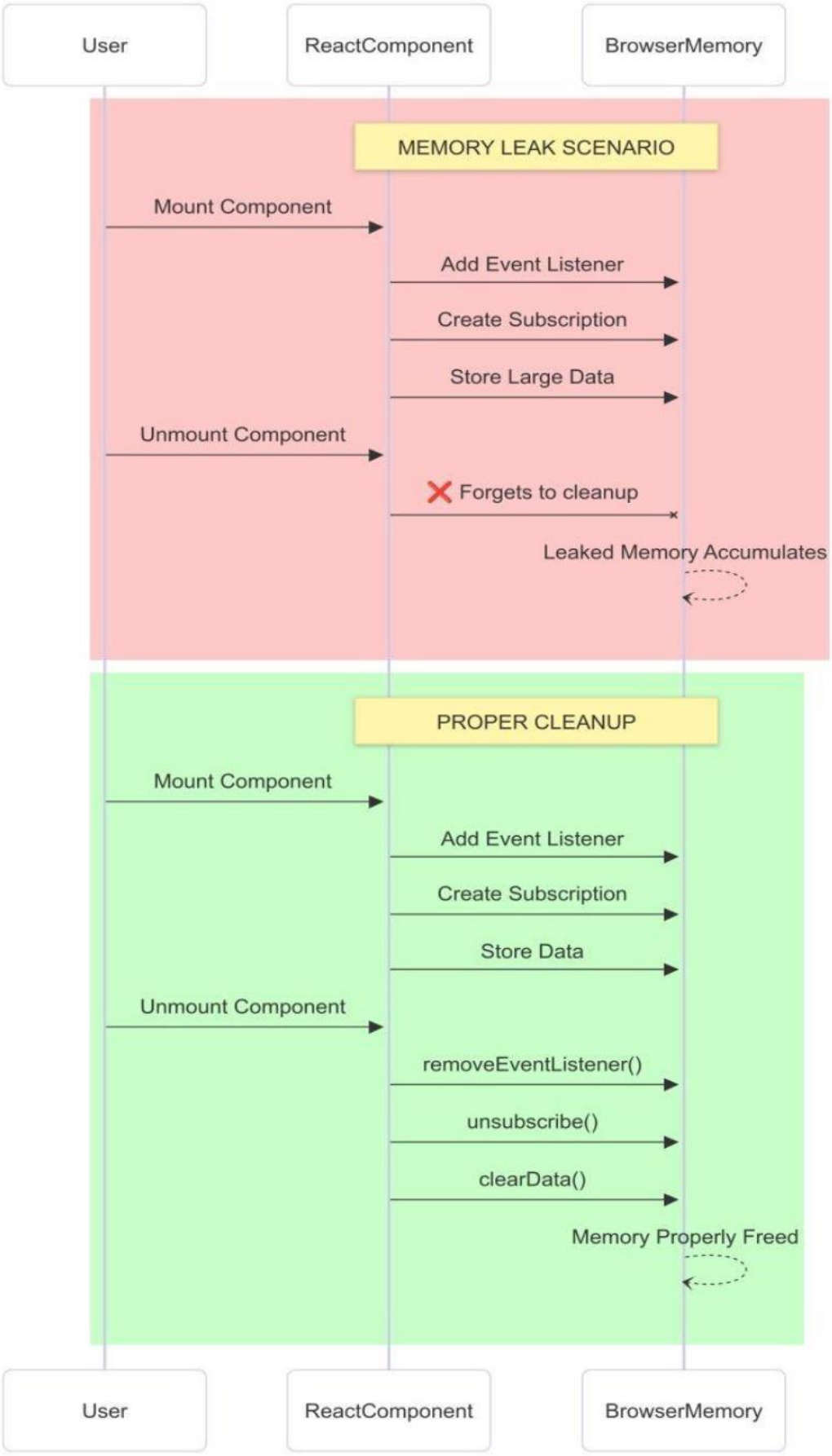
- Open your app in Chrome.
- Go to **Performance → Record**.
- Interact with your app (navigate, mount/unmount components).
- Stop recording and **look for memory spikes** that never drop.
    - If memory keeps growing, that's a leak.

Use Chrome DevTools → Memory Tab

- Go to **Memory** tab → select **Heap Snapshot**.
- Take a snapshot → interact with the app → take another snapshot.
- Compare them.
- Look for:
- Detached DOM nodes (meaning something references removed components)
- React component instances not being garbage-collected.

Use React Developer Tools Profiler

- Record interactions.
- Check which components re-render unexpectedly or stay mounted longer than expected.
- Often reveals components keeping subscriptions, refs, or large data in memory.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ReactComponent
BrowserMemory
MEMORY LEAK SCENARIO
Mount Component
Add Event Listener
Create Subscription
Store Large Data
Unmount Component
Forgets to cleanup
Leaked Memory Accumulates
PROPER CLEANUP
Mount Component
Add Event Listener
Create Subscription
Store Data
Unmount Component
removeEventListener()
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User
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